

Supplementary Material for “IMBANG: Balancing DDoS Mitigation and Service Outages with Safety-Spanning Group-Relative Policy Optimisation”

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S1. OVERVIEW

This document supports the letter with material that does not fit in four pages. It contains:

- the complete simulator, service model and symbolic shield that the letter adopts from Sentinel [1] (Sections S2 and S3);
- full proofs of Propositions 1 and 2, and an empirical check of Proposition 2 over whole episodes (Section S4);
- every training and evaluation setting (Sections S5 and S6);
- extended results: all metrics with confidence intervals, all trained methods, paired tests, the red team, per-seed results and rule distillation (Sections S7–S9);
- the simulated operator study behind the ICMP limitation (Section S10);
- instructions for the accompanying code and data archive (Section S11).

Tables, figures and equations here are numbered S1, S2, ...; plain numbers such as “Table II” refer to the letter. All results use seeds 42–51 and are paired with Sentinel’s ten released checkpoints [2]. Every number is generated from the per-seed files in the archive by `scripts/supp_assets.py`.

The method’s name, IMBANG, stands for **I**nterpretable, **M**onotone-safe, **B**udgeted, **A**dversary-free, **N**euro-symbolic **G**RPO. In the code and data files it is called `span`; Table S1 lists all code names.

TABLE S1
METHOD NAMES IN THE LETTER AND IN THE CODE

Name in the letter	Code name	Description
IMBANG	<code>span</code>	safe-side member + outage budget + MaxMC curriculum
C-GRPO	<code>span_cgrpo</code>	outage budget with channel-wise cost normalisation [3], no safe-side member
IMBANG + C-GRPO norm.	<code>span_cgrpo_safe</code>	IMBANG with channel-wise cost normalisation
GRPO + budget	<code>span_nosafe</code>	IMBANG without the safe-side member
+ entropy bonus $\times 6$	<code>span_entropy</code>	GRPO + budget with six times the entropy weight
+ uncentred cost	<code>span_abcscost</code>	GRPO + budget with a cost advantage referenced to κ instead of the group mean
MaxMC curriculum, no budget	<code>maxmc_only</code>	curriculum only (extra baseline)
GRPO, no budget	<code>nscsd</code>	GRPO with archive curriculum, no budget
Plain GRPO	<code>grpo_vanilla</code>	GRPO without archive, elites, anchor or shield tuning (Fig. 2 only)
PPO + shield	<code>ppo</code>	PPO with a critic, same shield and simulator budget
Sentinel	–	released <code>best_benchmark</code> checkpoints of [1]

S2. SIMULATOR AND SERVICE MODEL

The letter uses Sentinel’s single-link simulator unchanged [1], [2]. We re-implemented it in batched form. Two tests in the archive (`tests/test_env_equivalence.py`) check that the port matches the reference code to within 10^{-5} . Table S3 shows that Sentinel’s released checkpoints reproduce its published results on our port. This section restates the model so that the proofs in Section S4 can be checked line by line.

Traffic. The link capacity is $C = 1000$ units per window. Benign traffic is $L_t \sim \text{Poisson}(300)$; during a flash crowd it is multiplied by a factor drawn from $\mathcal{U}(2.5, 4)$. The attack has a family $k_t \in \{\text{none}, \text{SYN}, \text{UDP}, \text{HTTP}, \text{ICMP}, \text{mixed}\}$, an intensity $i_t \in [0, 1]$ and a mutation level $m_t \in [0, 1]$. Its volume is $A_t = 1.5 C i_t$, so a full-intensity flood alone exceeds the link by half.

Action. The defender chooses a load threshold $\tau_t \in (0, 1)$, a filtering focus $f_t \in \{\text{SrcIP}, \text{Proto}, \text{Size}\}$ and a drop probability $d_t \in (0, 1)$. Only the load above the threshold is filtered. The share of the offered load exposed to filtering is

$$\alpha_t(\tau) = \text{clip}\left(\frac{L_t + A_t - C\tau}{L_t + A_t}, 0, 1\right). \quad (\text{S1})$$

Effectiveness and collateral damage. The match between focus and family is $\eta_t = \eta_{k_t f_t} (1 - m_t/2)$, with the base values of Table S2. In an attacked window the shares of attack traffic dropped and of benign traffic lost are

$$\text{eff}_t = \min\{1, \eta_t d_t \alpha_t(\tau_t)\}, \quad (\text{S2})$$

$$\text{CD}_t = \min\left\{1, 0.4 d_t (1 - \eta_t) \alpha_t(\tau_t) + 0.2 [L_t - C\tau_t]^+ / L_t\right\}. \quad (\text{S3})$$

Without an attack, the collateral damage is $\min\{1, 0.05 d_t \alpha_t(\tau_t) + 0.1 [L_t - C\tau_t]^+ / L_t\}$. The leakage is $\text{leak}_t = 1 - \text{eff}_t$ in attacked windows.

Service quality. With the congestion factor $\rho_t = \min\{1, C/(L_t + A_t)\}$, computed on the offered (pre-filter) load,

$$\text{SQ}_t = \rho_t (1 - \text{CD}_t). \quad (\text{S4})$$

A window is a *severe outage* if $\text{SQ}_t < 0.5$ and *degraded* if $\text{SQ}_t < 0.75$. Under a full-intensity flood, $\rho_t \approx 1000/1800 = 0.556$, so even perfect filtering cannot lift service quality above 0.556, and a collateral damage of only about 0.10 already causes a severe outage.

Score. The operator’s per-window composite score is Sentinel’s benchmark,

$$b_t = \text{SQ}_t - \text{leak}_t - 0.5 \mathbf{1}[\text{SQ}_t < 0.5] - 0.25 \mathbf{1}[\text{SQ}_t < 0.75]. \quad (\text{S5})$$

All GRPO variants and the PPO baseline optimise b_t directly. Sentinel’s checkpoints were trained on its shaped reward and selected with b_t .

Observation. The defender observes twelve aggregate indicators of the served traffic in $[0, 1]$: packet rate, byte rate, link utilisation, the TCP, UDP, ICMP and HTTP shares, the entropies of source address, destination port and packet size, a flow count and a queue (overload) signal. Small Gaussian noise is added to the protocol shares, the entropies and the packet rate. The protocol shares and entropies depend on the share of attack traffic that is served. The observation therefore depends on the previous action, which matters for the multi-step discussion in Section S4.

TABLE S2
BASE MATCH EFFECTIVENESS η_{kf} BETWEEN ATTACK FAMILY AND FILTERING FOCUS [2]

Family	SrcIP	Proto	Size
SYN	0.6	0.8	0.0
UDP	0.0	0.9	0.0
HTTP	0.7	0.0	0.5
ICMP	0.0	0.9	0.0
Mixed	0.4	0.4	0.4

TABLE S3
SENTINEL’S RELEASED CHECKPOINTS: PUBLISHED RESULTS AND OUR PORT

Scenario	Quality		Leakage		Severe / 500	
	Published	Port	Published	Port	Published	Port
Benign	1.000	1.000	0.000	0.000	0.0	0.0
Chaos/flash crowd	0.699	0.701	0.836	0.836	25.3	22.0
Mild	0.971	0.971	0.921	0.921	0.0	0.0
Strong	0.519	0.519	0.815	0.815	75.0	78.2
Zero-shot ICMP	0.529	0.529	0.747	0.747	0.0	0.0

Published: Sentinel’s evaluation tables. Port: our batched simulator with Sentinel’s evaluation protocol and its released seed-42–51 checkpoints.

S3. SYMBOLIC SHIELD

The shield $\mathcal{S}_\phi$ is Sentinel’s rule set [1], applied in the order of Table S4. Each rule’s trigger depends only on the observation $\mathbf{o}_t$ and on the proposed focus f . Each rule that changes d or τ is a clamp of the form $\min\{\cdot, c\}$ or $\max\{\cdot, c\}$ with a constant c . IMBANG may tune the three PanicGuard constants $\phi = (g_u, g_d, g_\tau)$, defaults $(0.6, 0.1, 0.4)$, under the certificate described in Section S5. All other constants stay as in Sentinel.

TABLE S4
SHIELD RULES IN ORDER OF APPLICATION (UTIL: LINK UTILISATION; SRC, SIZE: ENTROPIES; UDP, ICMP, HTTP: PROTOCOL SHARES)

#	Rule	Trigger	Repair
1	PanicGuard (drop)	$\text{util} < g_u$	$d \leftarrow \min\{d, g_d\}$
2	PanicGuard (threshold)	$\text{util} < g_u$	$\tau \leftarrow \max\{\tau, g_\tau\}$
3	HTTP source-address exception	$\text{src} > 0.85, f = \text{SrcIP}, \text{util} > 0.6, \text{http} > 0.3$	keep f
4	SpoofingGuard (protocol)	$\text{src} > 0.85, f = \text{SrcIP}, \text{util} > 0.6$, rule 3 not met, a protocol share > 0.5	$f \leftarrow \text{Proto}$
5	SpoofingGuard (size)	as rule 4 but no protocol share > 0.5 and size < 0.3	$f \leftarrow \text{Size}$
6	SpoofingGuard (fallback)	as rule 4 but no protocol share > 0.5 and size ≥ 0.3	$f \leftarrow \text{Proto}$
7	Protocol-anomaly guard	a protocol share > 0.8 , util > 0.6 , $f \neq \text{Proto}$, not ($\text{http} > 0.8$ and $f = \text{SrcIP}$)	$f \leftarrow \text{Proto}$
8	ICMP focus override	$\text{icmp} > 0.3, \text{util} > 0.6, f \neq \text{Proto}$	$f \leftarrow \text{Proto}$
9	ICMP drop override	$\text{icmp} > 0.3, \text{util} > 0.6$	$d \leftarrow \max\{d, 0.5\}$
10	ICMP threshold override	$\text{icmp} > 0.3, \text{util} > 0.6$	$\tau \leftarrow \min\{\tau, 0.5\}$
11	Extreme-drop cap	always	$d \leftarrow \min\{d, 0.95\}$

S4. PROOFS AND AN EMPIRICAL CHECK

Setting. A GRPO group contains G members that start from the same state and replay the same traffic, so they differ only in their actions. Member g collects the objective $O_g = \sum_{h=1}^H b_h$ and the outage count $K_g = \sum_{h=1}^H \mathbf{1}[\text{SQ}_h < 0.5]$. With a multiplier $\lambda \geq 0$, its return is $R_g = O_g - \lambda K_g$. The group-normalised advantage is

$$\hat{A}_g = \frac{R_g - \bar{R}}{s_R}, \quad \bar{R} = \frac{1}{G} \sum_{g'} R_{g'}, \quad s_R = \left(\frac{1}{G} \sum_{g'} (R_{g'} - \bar{R})^2 \right)^{1/2}, \quad (\text{S6})$$

with $\hat{A}_g := 0$ when $s_R = 0$.

Proposition 1 (Constraint cancellation). *If $K_g \equiv K$ for every member of a group, then $\hat{A}_g$ does not depend on λ . The same holds for the channel-wise advantage $\hat{A}_g = z(O)_g - \lambda z(K)_g$ of Constrained GRPO [3], with the convention $z(K)_g := 0$ when $s_K = 0$.*

Proof. If $K_g \equiv K$, then $\bar{R} = \bar{O} - \lambda K$, so $R_g - \bar{R} = O_g - \bar{O}$ for every g . The two sets of deviations coincide, so $s_R = s_O$ and $\hat{A}_g = (O_g - \bar{O})/s_O$, which does not contain λ . For the channel-wise form, $K_g \equiv K$ gives $s_K = 0$, hence $z(K) \equiv 0$ and $\hat{A}_g = z(O)_g$. $\square$

Remark 1. *The argument uses only the fact that the advantage is invariant to adding the same constant to every return in a group. It therefore covers any group-centred advantage, with or without scaling. In such a group, the policy-gradient estimate is identical for every value of λ , so raising the multiplier cannot change the update. This is what Fig. 4 of the letter shows for “budget only”. An uncentred cost term, $\hat{A}_g - \lambda(K_g/H - \kappa)$, survives the centring but adds the same value to every member of such a group. In the clipped surrogate this shifts all sampled actions alike instead of pointing towards safer ones. This is consistent with the result in Table S12 that the uncentred cost does not restore feasibility.*

Proposition 2 (Safe-side direction). *Let a policy member and the safe-side member observe the same $\mathbf{o}_t$ and face the same traffic in window t . The policy member proposes (τ, f, d) , and the safe-side member executes $(\tilde{\tau}, \tilde{f}, \tilde{d}) = (\tau + (1-u)(1-\tau), f, u d)$ with $u \in [0.2, 0.8]$. Then, after shielding, $\text{CD}_t^{\text{safe}} \leq \text{CD}_t^{\text{policy}}$ and $\text{SQ}_t^{\text{safe}} \geq \text{SQ}_t^{\text{policy}}$. Consequently, the safe-side member is in a severe (or degraded) outage only if the policy member is.*

Proof. (i) *Before the shield.* Since $0 \leq u \leq 1$ and $\tau \leq 1$, we have $\tilde{d} = u d \leq d$ and $\tilde{\tau} = \tau + (1-u)(1-\tau) \geq \tau$. Both stay in $[0, 1]$.

(ii) *Through the shield.* The rules that change the focus (3–8 in Table S4) are triggered by $\mathbf{o}_t$ and the proposed focus only, and $\tilde{f} = f$. Both members therefore leave the shield with the same focus f' . Each rule acting on d (1, 9, 11) has a trigger that depends on $\mathbf{o}_t$ only and applies $\min\{\cdot, c\}$ or $\max\{\cdot, c\}$. These maps are non-decreasing, and so is their composition. Hence $\tilde{d} \leq d$ implies $\mathcal{S}(\tilde{d}) \leq \mathcal{S}(d)$. The same argument applies to τ (rules 2 and 10) and gives $\mathcal{S}(\tilde{\tau}) \geq \mathcal{S}(\tau)$.

(iii) *Through the service model.* With the same family, mutation and final focus f' , both members have the same η_t . In (S3), $1 - \eta_t \geq 0$; $\alpha_t(\tau)$ and $[L_t - C\tau]^+$ are non-increasing in τ ; and the expression is non-decreasing in d . The no-attack expression has the same monotonicity, and $\min\{1, \cdot\}$ preserves it. Hence $\text{CD}_t^{\text{safe}} \leq \text{CD}_t^{\text{policy}}$. Finally, ρ_t in (S4) depends on the offered load only, not on the action, so $\text{SQ}_t^{\text{safe}} = \rho_t(1 - \text{CD}_t^{\text{safe}}) \geq \rho_t(1 - \text{CD}_t^{\text{policy}}) = \text{SQ}_t^{\text{policy}}$. Both outage indicators are monotone in SQ_t . $\square$

Scope. Proposition 2 is a statement about a single window, given the same observation. Two assumptions are structural. First, congestion is computed on the offered load. If filtering took place upstream of the bottleneck, a weaker filter could itself overload the link. Second, the shield acts through coordinate-wise clamps whose triggers do not depend on d or τ .

Over a rollout the two members’ observations can diverge, because the protocol shares and entropies depend on how much attack traffic was served. A weaker filter serves more attack traffic, which can change later observations and hence later actions. The single-window guarantee therefore does not, by itself, imply an ordering of whole episodes.

We test the multi-step version empirically in Table S5. Each trained IMBANG policy and its safe-shifted copy, with u drawn once per episode, face identical traffic in all nine Protocol B scenarios. In all 1,800 episodes (ten seeds, nine scenarios, 20 episodes each), the safe copy had more severe windows than the policy in 0.0% of episodes, that is, never. Under heavy floods it usually had fewer, and its episode-average collateral damage never exceeded the policy’s. As the construction intends, it pays for this with higher leakage, which is why it serves as a training signal rather than as a deployed policy.

TABLE S5
MULTI-STEP CHECK OF THE SAFE-SIDE DIRECTION (IMBANG, TEN SEEDS, 20 EPISODES OF 100 WINDOWS)

Scenario	Severe / 500		Collateral damage		Leakage		Episodes (%)	
	Policy	Safe	Policy	Safe	Policy	Safe	Safe fewer	Safe more
Benign	0.00	0.00	0.0000	0.0000	0.000	0.000	0.0	0.0
Mild	0.00	0.00	0.0187	0.0012	0.941	0.995	0.0	0.0
Strong	38.88	0.78	0.0633	0.0272	0.814	0.922	85.0	0.0
Chaos/flash	10.72	8.00	0.0386	0.0138	0.870	0.953	36.5	0.0
ICMP (zero-shot)	18.00	0.45	0.0541	0.0465	0.716	0.757	41.0	0.0
Pulsing*	16.05	0.40	0.0269	0.0123	0.847	0.933	73.5	0.0
Polymorphic*	14.23	0.38	0.0785	0.0308	0.890	0.957	93.0	0.0
Boundary*	0.00	0.00	0.0576	0.0142	0.835	0.959	0.0	0.0
ICMP-chaos*	20.88	17.48	0.0480	0.0283	0.800	0.873	34.5	0.0

Each trained IMBANG policy and its safe-shifted copy ($\tilde{d} = u d$, $\tilde{\tau} = \tau + (1 - u)(1 - \tau)$, $u \sim \mathcal{U}(0.2, 0.8)$ per episode) face identical traffic with the seeds of Protocol B; the policy columns therefore equal Table S8. Safe fewer / more: episodes in which the safe copy has strictly fewer / more severe windows.

S5. TRAINING DETAILS

Groups. Each iteration samples $B = 32$ training contexts, i.e. attack levels with their traffic. Each context is run for a random warm-up of 0–40 windows under the current policy and then forked into $G = 8$ members that share all random numbers from the fork onwards [4]. In IMBANG the members are:

- four ordinary policy samples;
- one safe-side member: a policy sample transformed as in Proposition 2, with $u \sim \mathcal{U}(0.2, 0.8)$ drawn once per rollout;
- two shield probes: policy samples deployed under a perturbed PanicGuard;
- one archive elite: the best stored solution for the level’s niche, replaced by a policy sample when none exists.

Each member is rolled out for $H = 8$ windows.

Update. The returns are $R_g = O_g - \lambda K_g$ and the advantages are group-normalised. Policy members and shield probes enter the clipped surrogate [5]. The safe-side member and the elite enter only through the term

$$-c [\hat{A}_g]^+ \log \pi_\theta(a_g | \mathbf{o}), \quad c = 0.5, \quad (\text{S7})$$

so the policy moves towards them only when they outperformed their group. The loss also contains an entropy bonus and a KL penalty towards the best validated checkpoint (the anchor). It is minimised with Adam for four epochs per iteration. Table S6 lists all values.

Budget and selection. Every 20 iterations the deterministic policy is evaluated on the validation battery (benign, mild, strong and chaos traffic; ten episodes of 100 windows each, with fixed random numbers). This gives the severe-outage rate $\hat{\sigma}$ and the validation score. The multiplier follows dual ascent, $\lambda \leftarrow [\lambda + \eta_\lambda(\hat{\sigma} - \kappa)]^+$, with $\eta_\lambda = 4$ and $\kappa = 0.05$. The deployed checkpoint is chosen lexicographically: any checkpoint that meets the budget beats every checkpoint that does not. Among feasible checkpoints the validation score decides; among infeasible ones, the lower outage rate does.

Curriculum. Training contexts come from 360 discrete levels: five family patterns (SYN, UDP, HTTP, mixed, switching), six intensity bands, three mutation bands, pulsing on or off, and flash crowds on or off. ICMP is excluded. At every validation round, each level is scored with the current policy (two episodes of 50 windows, fixed random numbers). Its MaxMC regret is the best score any earlier snapshot achieved on that level minus the current score [6]. With probability 0.5 a context is drawn from the prioritised-replay distribution of [7]: rank-based regret priority with temperature 0.3, mixed with a 0.3 share of staleness. Otherwise it is drawn uniformly over the levels.

Certified shield tuning. Every 20 iterations, eight PanicGuard candidates are proposed. Up to three lie along the direction accumulated from the shield probes’ advantages; the rest are random perturbations. A candidate is adopted only if it meets three conditions:

- it improves the validation score by at least 0.002;
- it keeps the validation outage rate within the budget;
- it shows no safety regression against Sentinel’s default shield on a certification battery of benign and flash-crowd traffic (ten episodes of 100 windows, fixed random numbers): service quality may not fall by more than 0.002 (benign) or 0.005 (flash crowd), and there may be no additional severe or degraded windows.

The certificate is empirical, not a formal guarantee.

TABLE S6
HYPERPARAMETERS (FIXED BEFORE THE TEN-SEED RUNS)

Setting	Value	Note
<i>GRPO learner (shared by all GRPO variants)</i>		
Iterations	500	validation every 20 iterations
Contexts per iteration B / group size G / horizon H	32 / 8 / 8	warm-up 0–40 windows before the fork
Group composition (IMBANG)	4 + 1 + 2 + 1	policy samples, safe-side member, shield probes, archive elite
Optimiser	Adam, lr 0.0003	4 epochs, minibatch 512, gradient-norm clip 0.5
Clipping / entropy / KL-anchor weight	0.2 / 0.005 / 0.02	KL to the best validated checkpoint
Weight of the likelihood term c	0.5	safe-side member and elite, $-c [\hat{A}]^+ \log \pi_\theta$
Objective	composite score b_t	Sentinel’s benchmark weights (severe 0.5, degraded 0.25)
<i>IMBANG-specific</i>		
Safe-side shift u	$\mathcal{U}(0.2, 0.8)$	drawn once per rollout
Outage budget κ	0.05	severe-outage rate on the validation battery
Dual step η_λ	4.0	per validation round, $\lambda_0 = 0$
Champion selection	feasibility first	then composite validation score
Curriculum	MaxMC, 360 levels	5 families $\times$ 6 intensity bands $\times$ 3 mutation bands $\times$ pulse $\times$ flash
Shield probes / perturbation	2 / $\sigma = 0.08$	certified PanicGuard tuning
<i>Network and evaluation</i>		
Actor	12–64–64 MLP	Beta heads for τ and d , categorical head for f (Sentinel’s architecture)
Seeds	42–51	paired with Sentinel’s released checkpoints
Validation battery	benign, mild, strong, chaos	ICMP never seen before testing
Protocol B / A	20 / 5 episodes	100 windows each
Protocol C (red team)	CEM, 48 $\times$ 8	4 episodes of 60 windows per candidate; worst scenario re-scored on 20 fresh episodes of 100

S6. EVALUATION PROTOCOLS AND STATISTICS

Protocol B (attacker-agnostic). Nine open-loop scenarios (Table S7), each with 20 episodes of 100 windows per seed. The first five mirror Sentinel’s evaluation scenarios. The last four are held-out stress tests that were never used for training or model selection. ICMP traffic appears only at test time.

Protocol A (Sentinel’s own attackers). For each seed, the attacker that Sentinel trained alongside its defender drives five scenarios with five episodes of 100 windows each, reproducing Sentinel’s evaluation script. Averages in the letter and in Table S11 are taken over the four attack scenarios and exclude benign traffic.

Protocol C (red team). A cross-entropy search explores the full 14-dimensional scenario space, ICMP included:

- eight iterations of 48 candidates;
- each candidate scored on four episodes of 60 windows;
- the elite is the best 20%.

The worst scenario found is then re-scored on 20 fresh episodes of 100 windows.

Common random numbers and statistics. For a given seed and scenario, all defenders face identical traffic, so comparisons are paired by seed. We report means with 95% t -intervals over the ten seeds. The tests are paired t -tests with Holm correction across the eight attack scenarios for each metric, and two-sided Wilcoxon signed-rank tests. With ten pairs, the smallest attainable Wilcoxon p -value is 0.002.

S7. EXTENDED RESULTS

Tables S8 and S9 extend Table II of the letter with confidence intervals, service quality, leakage and the PPO baseline. Table S10 gives the paired tests behind the significance marks. Table S11 reports Protocol A by scenario, and Table S12 compares all trained methods across the three protocols. Fig. S1 shows the seed-level pairs: IMBANG scores higher than Sentinel on every seed and has a far narrower spread of outages. Fig. S2 breaks the severe outages down by scenario.

Table S12 also lists the four reference heuristics shipped with Sentinel’s code. They show why the outage count alone is not a sufficient measure. Sentinel’s adaptive heuristic matches IMBANG’s outage count (14.5 against 14.8 per 500 windows), but only because it filters very little: it lets 96.9% of the attack traffic through, against 83.9% for IMBANG. Its composite score is lower than that of every learned defender.

TABLE S7
PROTOCOL B EVALUATION SCENARIOS

Scenario	Attack families (weights)	Persistence	Intensity	Mutation	Pulse period / duty	Flash prob.	Role
Benign	none	0.95	[0.0, 0.0]	[0.0, 1.0]	–	0.00	Sentinel scenario
Mild	SYN, UDP, HTTP, mixed	0.95	[0.0, 0.3]	[0.0, 1.0]	–	0.00	Sentinel scenario
Strong	SYN, UDP, HTTP, mixed	0.95	[1.0, 1.0]	[0.0, 1.0]	–	0.00	Sentinel scenario
Chaos/flash	none, SYN, UDP, HTTP, mixed	0.95	[0.0, 1.0]	[0.0, 1.0]	–	0.05	Sentinel scenario
ICMP (zero-shot)	ICMP	0.95	[1.0, 1.0]	[0.0, 1.0]	–	0.00	Sentinel scenario (unseen family)
Pulsing*	SYN, UDP, HTTP, mixed	0.95	[1.0, 1.0]	[0.0, 1.0]	10 / 0.5	0.00	held-out stress test
Polymorphic*	SYN, UDP, HTTP, mixed	0.50	[0.5, 1.0]	[0.7, 1.0]	–	0.00	held-out stress test
Boundary*	SYN, UDP, HTTP, mixed	0.95	[0.3, 0.6]	[0.0, 1.0]	–	0.00	held-out stress test
ICMP-chaos*	none (0.12), SYN (0.12), UDP (0.12), HTTP (0.12), ICMP (0.38), mixed (0.12)	0.95	[0.5, 1.0]	[0.0, 1.0]	–	0.05	held-out stress test

Families switch with probability 1—persistence per window; intensity and mutation are redrawn uniformly from their ranges on each switch. *Never used for training or model selection.

TABLE S8
PROTOCOL B: COMPOSITE SCORE AND SEVERE OUTAGES PER 500 WINDOWS (MEAN $\pm$ 95% CI OVER TEN SEEDS)

Scenario	Composite score $\uparrow$				Severe outages $\downarrow$			
	Sentinel	PPO	C-GRPO	IMBANG	Sentinel	PPO	C-GRPO	IMBANG
Benign	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Mild	0.009 $\pm$ 0.003	0.008 $\pm$ 0.003	0.053 $\pm$ 0.014	0.040 $\pm$ 0.006	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Strong	−0.746 $\pm$ 0.066	−0.647 $\pm$ 0.038	−0.612 $\pm$ 0.048	−0.583 $\pm$ 0.018	155.2 $\pm$ 49.9	83.4 $\pm$ 37.8	85.9 $\pm$ 59.9	38.9 $\pm$ 14.6
Chaos/flash	−0.169 $\pm$ 0.019	−0.150 $\pm$ 0.021	−0.114 $\pm$ 0.018	−0.123 $\pm$ 0.020	14.1 $\pm$ 2.3	10.6 $\pm$ 1.4	16.8 $\pm$ 9.9	10.7 $\pm$ 1.2
ICMP (zero-shot)	−0.474 $\pm$ 0.010	−0.467 $\pm$ 0.012	−0.474 $\pm$ 0.008	−0.458 $\pm$ 0.020	4.6 $\pm$ 9.3	3.7 $\pm$ 4.1	23.3 $\pm$ 35.5	18.0 $\pm$ 17.3
Pulsing*	−0.303 $\pm$ 0.026	−0.250 $\pm$ 0.008	−0.228 $\pm$ 0.017	−0.225 $\pm$ 0.009	60.5 $\pm$ 20.1	27.9 $\pm$ 10.0	37.7 $\pm$ 27.2	16.1 $\pm$ 5.8
Polymorphic*	−0.454 $\pm$ 0.008	−0.442 $\pm$ 0.009	−0.442 $\pm$ 0.014	−0.430 $\pm$ 0.008	14.9 $\pm$ 6.1	10.5 $\pm$ 3.6	24.0 $\pm$ 12.9	14.2 $\pm$ 3.4
Boundary*	0.003 $\pm$ 0.012	0.017 $\pm$ 0.005	0.084 $\pm$ 0.018	0.071 $\pm$ 0.018	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
ICMP-chaos*	−0.299 $\pm$ 0.024	−0.276 $\pm$ 0.022	−0.255 $\pm$ 0.030	−0.257 $\pm$ 0.022	25.4 $\pm$ 4.1	21.1 $\pm$ 2.4	25.2 $\pm$ 5.2	20.9 $\pm$ 1.8
Average (attacks)	−0.304 $\pm$ 0.014	−0.276 $\pm$ 0.006	−0.248 $\pm$ 0.009	−0.246 $\pm$ 0.009	34.4 $\pm$ 9.5	19.6 $\pm$ 6.8	26.6 $\pm$ 17.8	14.8 $\pm$ 3.3

*Held-out stress test. Severe outages are windows with service quality below 0.5. PPO: PPO + shield trained with the same budget of simulator windows.

TABLE S9
PROTOCOL B: SERVICE QUALITY AND ATTACK LEAKAGE (MEAN $\pm$ 95% CI OVER TEN SEEDS)

Scenario	Service quality $\uparrow$				Attack leakage $\downarrow$			
	Sentinel	PPO	C-GRPO	IMBANG	Sentinel	PPO	C-GRPO	IMBANG
Benign	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
Mild	0.991 $\pm$ 0.002	0.994 $\pm$ 0.002	0.970 $\pm$ 0.004	0.981 $\pm$ 0.004	0.983 $\pm$ 0.004	0.986 $\pm$ 0.004	0.917 $\pm$ 0.017	0.941 $\pm$ 0.009
Strong	0.511 $\pm$ 0.005	0.516 $\pm$ 0.003	0.513 $\pm$ 0.006	0.520 $\pm$ 0.002	0.852 $\pm$ 0.025	0.829 $\pm$ 0.011	0.789 $\pm$ 0.027	0.814 $\pm$ 0.018
Chaos/flash	0.837 $\pm$ 0.011	0.842 $\pm$ 0.011	0.829 $\pm$ 0.012	0.838 $\pm$ 0.012	0.908 $\pm$ 0.017	0.899 $\pm$ 0.016	0.843 $\pm$ 0.014	0.870 $\pm$ 0.018
ICMP (zero-shot)	0.529 $\pm$ 0.002	0.528 $\pm$ 0.002	0.526 $\pm$ 0.004	0.526 $\pm$ 0.003	0.748 $\pm$ 0.019	0.742 $\pm$ 0.015	0.727 $\pm$ 0.035	0.716 $\pm$ 0.029
Pulsing*	0.759 $\pm$ 0.002	0.761 $\pm$ 0.001	0.760 $\pm$ 0.003	0.763 $\pm$ 0.001	0.877 $\pm$ 0.016	0.858 $\pm$ 0.010	0.825 $\pm$ 0.018	0.847 $\pm$ 0.009
Polymorphic*	0.657 $\pm$ 0.006	0.660 $\pm$ 0.006	0.647 $\pm$ 0.010	0.662 $\pm$ 0.004	0.905 $\pm$ 0.010	0.902 $\pm$ 0.007	0.868 $\pm$ 0.016	0.890 $\pm$ 0.008
Boundary*	0.914 $\pm$ 0.007	0.922 $\pm$ 0.006	0.893 $\pm$ 0.008	0.907 $\pm$ 0.007	0.908 $\pm$ 0.011	0.905 $\pm$ 0.006	0.807 $\pm$ 0.019	0.835 $\pm$ 0.021
ICMP-chaos*	0.710 $\pm$ 0.010	0.714 $\pm$ 0.010	0.710 $\pm$ 0.009	0.715 $\pm$ 0.009	0.829 $\pm$ 0.018	0.816 $\pm$ 0.014	0.783 $\pm$ 0.030	0.800 $\pm$ 0.018
Average (attacks)	0.738 $\pm$ 0.003	0.742 $\pm$ 0.002	0.731 $\pm$ 0.005	0.739 $\pm$ 0.003	0.876 $\pm$ 0.013	0.867 $\pm$ 0.008	0.820 $\pm$ 0.016	0.839 $\pm$ 0.013

*Held-out stress test. Severe outages are windows with service quality below 0.5. PPO: PPO + shield trained with the same budget of simulator windows.

Red team. Table S13 details the worst cases. For Sentinel the search most often settles on HTTP floods; for the PPO and GRPO-trained defenders it mostly settles on mixed floods. Among learned defenders, IMBANG and its variant with C-GRPO normalisation have the best worst cases (−0.72 and −0.71) and the smallest spread across seeds. Sentinel’s adaptive heuristic reaches a similar worst case (−0.71), but its average leakage is far higher (Table S12).

Operating curve. Table S14 lists the data of Fig. 3 of the letter.

S8. MECHANISM DIAGNOSTICS

Table S15 gives the per-method values behind Fig. 2 of the letter. Across methods, a narrow drop spread goes with a low span rate and with frequent outages: plain GRPO samples drop probabilities with a spread of only 0.05 and is in a severe outage in 76% of overload windows.

The table also shows that the safe-side member does not work by widening the trained policy’s own action spread. Under strong floods, the budget-only policies span 93% of overload groups yet remain in outage in 35% of windows; IMBANG spans

TABLE S10
PAIRED TESTS OVER TEN SEEDS (PROTOCOL B)

Scenario	Composite score				Severe outages / 500			
	Δ	Holm p	Wilcoxon p	d_z	Δ	Holm p	Wilcoxon p	d_z
<i>IMBANG vs Sentinel</i>								
Mild	+0.031	1.1×10^{-6}	2.0×10^{-3}	+4.65	+0.0	1.00	1.00	+0.00
Strong	+0.164	2.7×10^{-4}	2.0×10^{-3}	+2.20	-116.3	3.5×10^{-3}	3.9×10^{-3}	-1.71
Chaos/flash	+0.046	1.3×10^{-4}	2.0×10^{-3}	+2.55	-3.4	0.09	0.04	-0.94
ICMP (zero-shot)	+0.016	0.04	0.04	+0.78	+13.4	0.57	0.08	+0.51
Pulsing*	+0.078	5.2×10^{-4}	2.0×10^{-3}	+1.94	-44.5	5.9×10^{-3}	3.9×10^{-3}	-1.55
Polymorphic*	+0.024	2.1×10^{-4}	2.0×10^{-3}	+2.33	-0.6	1.00	0.98	-0.06
Boundary*	+0.068	6.1×10^{-5}	2.0×10^{-3}	+2.84	+0.0	1.00	1.00	+0.00
ICMP-chaos*	+0.042	5.9×10^{-3}	5.9×10^{-3}	+1.27	-4.6	0.12	0.02	-0.86
<i>IMBANG vs C-GRPO</i>								
Mild	-0.013	0.28	0.05	-0.78	+0.0	1.00	1.00	+0.00
Strong	+0.029	0.37	0.02	+0.62	-47.0	0.66	0.04	-0.62
Chaos/flash	-0.009	0.37	0.05	-0.64	-6.0	0.81	9.8×10^{-3}	-0.43
ICMP (zero-shot)	+0.016	0.35	0.11	+0.69	-5.3	1.00	0.65	-0.10
Pulsing*	+0.003	1.00	0.70	+0.13	-21.6	0.70	5.9×10^{-3}	-0.58
Polymorphic*	+0.012	0.32	3.9×10^{-3}	+0.73	-9.8	0.80	0.05	-0.48
Boundary*	-0.013	0.37	0.11	-0.63	+0.0	1.00	1.00	+0.00
ICMP-chaos*	-0.002	1.00	1.00	-0.06	-4.3	0.70	3.9×10^{-3}	-0.57

Δ : IMBANG minus comparator, averaged over seeds. Holm: paired t -test, Holm-corrected across the eight scenarios per metric. Wilcoxon: signed-rank test (two-sided, uncorrected; its smallest attainable value with ten pairs is 0.002). d_z : standardised paired effect size.

TABLE S11
PROTOCOL A: SENTINEL'S OWN TRAINED ATTACKERS (TEN SEEDS, FIVE EPISODES EACH)

Scenario	Composite $\uparrow$		Severe / 500 $\downarrow$		Leakage $\downarrow$		Quality $\uparrow$	
	Sentinel	IMBANG	Sentinel	IMBANG	Sentinel	IMBANG	Sentinel	IMBANG
Benign	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000	0.000	1.000	1.000
Mild	0.051 $\pm$ 0.035	0.141 $\pm$ 0.062	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.921	0.808	0.971	0.950
Strong	-0.625 $\pm$ 0.193	-0.509 $\pm$ 0.069	78.2 $\pm$ 123.0	5.5 $\pm$ 8.3	0.815	0.778	0.519	0.524
Chaos/flash crowd	-0.288 $\pm$ 0.150	-0.251 $\pm$ 0.144	22.0 $\pm$ 4.3	19.8 $\pm$ 5.8	0.836	0.811	0.701	0.711
Zero-shot ICMP	-0.468 $\pm$ 0.029	-0.439 $\pm$ 0.033	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.747	0.715	0.529	0.526
Average (attacks)	-0.332 $\pm$ 0.062	-0.265 $\pm$ 0.039	25.1 $\pm$ 30.5	6.3 $\pm$ 3.1	0.830	0.778	0.680	0.678

For each seed, the attacker that Sentinel co-trained with its defender drives the traffic, as in Sentinel's own evaluation script; for IMBANG this is a transfer test. Leakage is averaged over all windows. Paired t -tests on the attack average: composite $p = 0.019$, severe $p = 0.22$, leakage $p = 0.018$, quality $p = 0.65$.

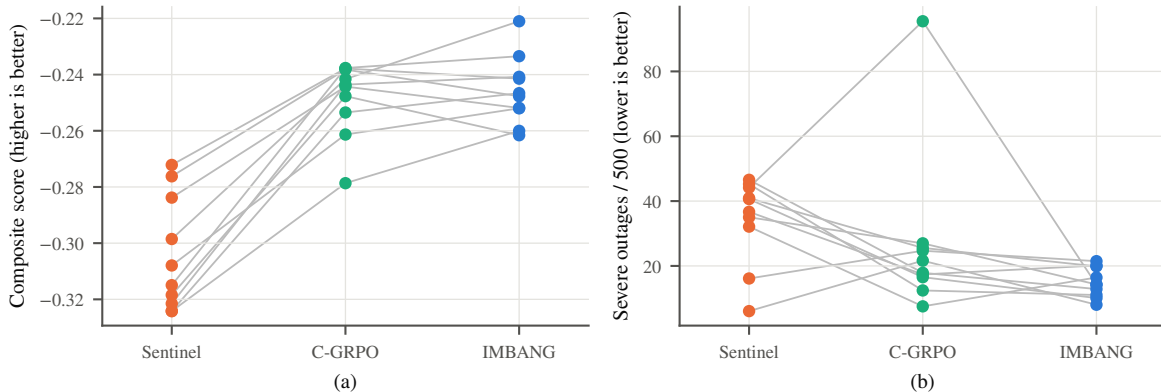


Fig. S1. Seed-level comparison under Protocol B (scenario average over the eight attack scenarios). Grey lines connect the same seed. (a) Composite score. (b) Severe outages per 500 windows.

TABLE S12
ALL TRAINED METHODS (TEN SEEDS; PROTOCOL B AVERAGED OVER THE EIGHT ATTACK SCENARIOS)

Method	Composite (B)	Severe (B)	Leakage (B)	Quality (B)	Composite (A)	Worst (C)	Budget met	Final λ	Span rate
Sentinel	-0.304 ± 0.014	34.4 ± 9.5	0.876	0.738	-0.332	-1.07 ± 0.17	–	–	–
PPO + shield	-0.276 ± 0.006	19.6 ± 6.8	0.867	0.742	-0.315	-0.79 ± 0.09	–	–	–
GRPO, no budget	-0.250 ± 0.009	91.5 ± 24.2	0.729	0.710	-0.276	-1.07 ± 0.08	–	–	–
MaxMC curriculum, no budget	-0.244 ± 0.010	52.4 ± 25.9	0.777	0.721	-0.246	-0.88 ± 0.12	0/10	0.00	0.42
GRPO + budget	-0.253 ± 0.011	36.5 ± 19.9	0.812	0.729	-0.248	-0.87 ± 0.15	1/10	7.69	0.43
+ entropy bonus $\times 6$	-0.249 ± 0.010	33.8 ± 19.4	0.810	0.729	-0.244	-0.85 ± 0.16	0/10	7.54	0.43
+ uncentred cost	-0.251 ± 0.010	34.7 ± 19.3	0.812	0.730	-0.242	-0.87 ± 0.13	1/10	7.38	0.44
C-GRPO	-0.248 ± 0.009	26.6 ± 17.8	0.820	0.731	-0.262	-0.80 ± 0.13	3/10	4.67	0.36
IMBANG	-0.246 ± 0.009	14.8 ± 3.3	0.839	0.739	-0.265	-0.72 ± 0.03	9/10	0.43	0.42
IMBANG + C-GRPO norm.	-0.247 ± 0.007	12.1 ± 2.5	0.843	0.739	-0.266	-0.71 ± 0.02	7/10	0.41	0.42
<i>Reference heuristics from Sentinel's code (no learning)</i>									
Adaptive heuristic	-0.342 ± 0.004	14.5 ± 0.9	0.969	0.766	-0.431	-0.71 ± 0.01	–	–	–
Static rule	-0.319 ± 0.005	24.8 ± 2.5	0.916	0.751	-0.385	-0.88 ± 0.09	–	–	–
Random actions + shield	-0.388 ± 0.004	86.1 ± 1.4	0.875	0.714	-0.471	-0.93 ± 0.02	–	–	–
Random actions	-0.476 ± 0.004	109.8 ± 1.3	0.920	0.698	-0.610	-0.97 ± 0.03	–	–	–

Worst (C): composite score of the worst scenario found by the red team. Budget met: seeds whose final validation outage rate is at most $\kappa = 0.05$. Span rate: share of training groups whose members differ in outage count, averaged over the last 100 iterations. –: not applicable or not logged. Adaptive: Sentinel's traffic-responsive heuristic; Static: fixed protocol filtering (threshold 0.8, drop 0.5).

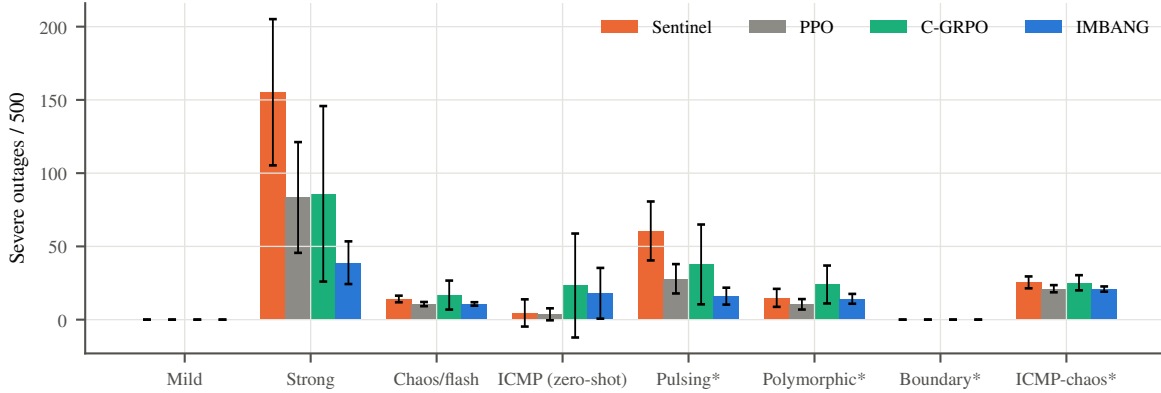


Fig. S2. Severe outages per 500 windows by Protocol B scenario (mean and 95% confidence interval over ten seeds; * marks held-out stress tests).

TABLE S13
PROTOCOL C: WORST CASE FOUND BY THE CROSS-ENTROPY RED TEAM (TEN SEEDS)

Method	Worst composite	Severe / 500	Quality	Leakage	Intensity	Worst family (seeds)
Sentinel	-1.07 ± 0.17	359.6 ± 127.3	0.494	0.960	0.98	http 7, mixed 3
PPO + shield	-0.79 ± 0.09	141.4 ± 84.4	0.505	0.909	0.97	mixed 9, syn 1
GRPO, no budget	-1.07 ± 0.08	466.6 ± 37.7	0.462	0.821	0.97	mixed 7, http 2, syn 1
MaxMC curriculum, no budget	-0.88 ± 0.12	254.3 ± 127.1	0.487	0.868	0.97	mixed 7, http 2, syn 1
GRPO + budget	-0.87 ± 0.15	213.0 ± 151.4	0.500	0.908	0.97	mixed 9, udp 1
+ entropy bonus $\times 6$	-0.85 ± 0.16	208.7 ± 161.1	0.501	0.893	0.97	mixed 9, syn 1
+ uncentred cost	-0.87 ± 0.13	214.3 ± 132.5	0.502	0.907	0.97	mixed 10
C-GRPO	-0.80 ± 0.13	155.5 ± 121.3	0.505	0.905	0.97	mixed 9, http 1
IMBANG	-0.72 ± 0.03	67.8 ± 46.3	0.521	0.921	0.96	mixed 8, http 1, udp 1
IMBANG + C-GRPO norm.	-0.71 ± 0.02	46.4 ± 24.3	0.523	0.935	0.97	mixed 8, http 2
Adaptive heuristic	-0.71 ± 0.01	29.4 ± 2.0	0.558	0.995	0.96	http 8, mixed 2
Static rule	-0.88 ± 0.09	150.1 ± 80.7	0.509	0.988	0.97	http 10
Random actions + shield	-0.93 ± 0.02	253.7 ± 15.0	0.486	0.919	0.97	http 6, mixed 4
Random actions	-0.97 ± 0.03	263.7 ± 16.6	0.479	0.935	0.97	udp 6, icmp 3, http 1

The red team searches the full 14-dimensional scenario space, ICMP included, for the scenario that minimises each defender's composite score.

TABLE S14
OPERATING CURVE OVER ATTACK INTENSITY (DATA OF FIG. 3 OF THE LETTER)

Intensity	Severe / 500 ↓			Leakage ↓			Quality ↑		
	Sentinel	C-GRPO	IMBANG	Sentinel	C-GRPO	IMBANG	Sentinel	C-GRPO	IMBANG
0.10	0.0	0.0	0.0	1.000	0.980	0.998	1.000	0.993	1.000
0.20	0.0	0.0	0.0	0.984	0.882	0.917	0.991	0.953	0.971
0.30	0.0	0.0	0.0	0.938	0.814	0.842	0.965	0.929	0.944
0.40	0.0	0.0	0.0	0.915	0.776	0.794	0.951	0.916	0.926
0.50	0.0	0.0	0.0	0.895	0.830	0.878	0.898	0.896	0.915
0.60	0.0	0.0	0.0	0.879	0.818	0.861	0.781	0.780	0.796
0.70	0.0	0.0	0.0	0.869	0.808	0.845	0.690	0.691	0.703
0.80	0.0	0.1	0.1	0.860	0.795	0.827	0.618	0.620	0.629
0.86	0.0	1.2	0.5	0.856	0.795	0.830	0.582	0.583	0.593
0.88	0.1	1.7	0.8	0.854	0.789	0.819	0.571	0.572	0.581
0.90	0.2	5.2	1.1	0.858	0.798	0.825	0.559	0.561	0.570
0.92	2.0	12.8	1.8	0.857	0.795	0.822	0.548	0.551	0.559
0.94	23.9	19.5	3.7	0.853	0.789	0.812	0.539	0.541	0.549
0.96	84.0	30.5	6.6	0.858	0.796	0.819	0.528	0.531	0.538
0.98	99.0	49.8	16.4	0.852	0.790	0.817	0.520	0.522	0.530
1.00	142.7	84.5	39.6	0.847	0.781	0.804	0.512	0.514	0.521

Training attack mix (SYN, UDP, HTTP, mixed), 20 episodes of 100 windows per point and seed, identical traffic for all defenders; means over ten seeds.

89% and is in outage in 19%. The same holds during training: Fig. S3 shows that the span rate, averaged over all training contexts, is similar with and without the safe-side member (0.42 and 0.43 over the last 100 iterations).

What the member adds is a group member that is guaranteed to be no less safe, and whose advantage is positive whenever the gentler action paid off. The update thus gets a direction towards fewer outages that does not depend on the policy’s own exploration. This is consistent with Fig. 4 of the letter, where the multiplier stays small for IMBANG but keeps growing without the member.

TABLE S15
CONSTRAINT-CANCELLATION PROBE ON THE 80 TRAINED CHECKPOINTS OF FIG. 2 OF THE LETTER

Method	Strong flood			Polymorphic flood		
	Span rate	Drop spread	Severe rate	Span rate	Drop spread	Severe rate
Sentinel	1.00 ± 0.00	0.215	0.45 ± 0.05	0.92 ± 0.03	0.216	0.10 ± 0.01
PPO + shield	1.00 ± 0.00	0.222	0.37 ± 0.04	0.87 ± 0.04	0.221	0.08 ± 0.01
Plain GRPO	0.47 ± 0.21	0.048	0.76 ± 0.08	0.62 ± 0.14	0.048	0.24 ± 0.05
GRPO, no budget	0.62 ± 0.20	0.079	0.63 ± 0.11	0.62 ± 0.15	0.077	0.18 ± 0.04
GRPO + budget	0.93 ± 0.05	0.161	0.35 ± 0.11	0.67 ± 0.13	0.160	0.08 ± 0.03
C-GRPO	0.78 ± 0.09	0.130	0.26 ± 0.11	0.49 ± 0.11	0.131	0.06 ± 0.03
IMBANG	0.89 ± 0.04	0.183	0.19 ± 0.03	0.60 ± 0.08	0.185	0.05 ± 0.01
IMBANG + C-GRPO norm.	0.85 ± 0.05	0.185	0.17 ± 0.03	0.57 ± 0.03	0.188	0.04 ± 0.01

From overload states reached after 20 deterministic windows, 64 contexts are forked into $G = 8$ stochastic members sharing traffic and rolled out for $H = 8$ windows. Span rate: share of groups whose members differ in outage count. Drop spread: within-group standard deviation of the sampled drop probability. Severe rate: share of member-windows in severe outage.

S9. PER-SEED RESULTS AND INTERPRETABILITY

Table S16 lists, for each seed, IMBANG’s Protocol B results, its final validation outage rate and multiplier, the fidelity of the distilled depth-4 rule trees, and the certified PanicGuard constants. Seed 43 is the one run whose final validation outage rate (0.076) exceeds the budget. The focus tree of seed 42 is reproduced below as an example of the rules an operator can audit. Class 0 is source-address filtering and class 1 is protocol filtering; the threshold and drop trees are in the archive.

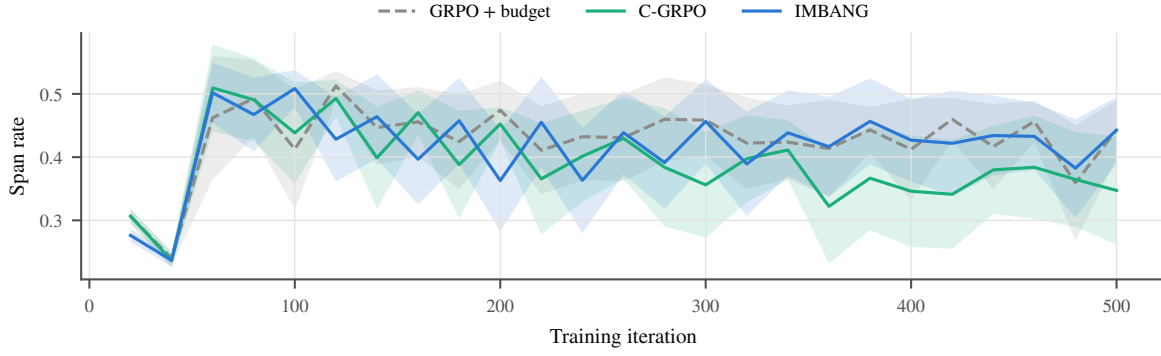


Fig. S3. Span rate during training: share of groups whose members differ in outage count, averaged over all training contexts (mean and 95% confidence band over ten seeds). The safe-side member does not raise this average; its effect is on the direction of the update (Section S8).

```

|--- http_ratio <= 0.219
|   |--- class: 1
|--- http_ratio > 0.219
|   |--- packet_size_entropy <= 0.644
|       |--- class: 0
|       |--- packet_size_entropy > 0.644
|           |--- udp_ratio <= 0.086
|               |--- total_packet_rate <= 0.785
|                   |--- class: 0
|                   |--- total_packet_rate > 0.785
|                       |--- class: 1
|                       |--- udp_ratio > 0.086
|                           |--- class: 1

```

TABLE S16
IMBANG PER SEED: RESULTS, BUDGET, RULE DISTILLATION AND CERTIFIED SHIELD

Seed	Composite (B)	Severe (B)	Val. outage rate	Budget met	Final λ	Focus acc. (%)	Threshold MAE	Drop MAE	Leaves	PanicGuard gate / cap / floor
42	-0.241	11.0	0.021	yes	0.00	99.8	0.011	0.022	33	0.52 / 0.07 / 0.42
43	-0.233	20.1	0.076	no	0.10	100.0	0.014	0.016	29	0.54 / 0.08 / 0.42
44	-0.252	21.5	0.033	yes	0.00	99.8	0.010	0.023	34	0.50 / 0.14 / 0.29
45	-0.252	14.3	0.011	yes	3.63	99.2	0.007	0.021	36	0.50 / 0.10 / 0.41
46	-0.241	12.9	0.049	yes	0.00	99.5	0.022	0.017	32	0.47 / 0.05 / 0.39
47	-0.260	14.2	0.031	yes	0.04	99.8	0.006	0.007	31	0.54 / 0.20 / 0.36
48	-0.247	19.9	0.025	yes	0.00	99.8	0.026	0.021	30	0.60 / 0.10 / 0.40
49	-0.221	16.4	0.028	yes	0.46	99.8	0.010	0.015	32	0.51 / 0.07 / 0.41
50	-0.262	10.1	0.007	yes	0.02	99.9	0.012	0.019	36	0.53 / 0.18 / 0.37
51	-0.248	8.0	0.010	yes	0.00	99.9	0.008	0.010	30	0.54 / 0.10 / 0.40

Rule distillation: depth-4 decision trees fitted to the final policy, evaluated on held-out states. PanicGuard defaults are 0.60 / 0.10 / 0.40.

S10. SIMULATED OPERATOR FOR ZERO-SHOT ICMP FLOODS

The letter notes that IMBANG has more severe outages than Sentinel under zero-shot ICMP floods, and that an operator applying Sentinel’s ICMP constants closes the gap. Table S17 gives the full study. It is a sensitivity analysis with a simulated operator, not a user study. The playbook constants are Sentinel’s own ICMP override values, fixed a priori. With 90% recognition accuracy the playbook lowers IMBANG’s ICMP outages from 17.9 to 2.1 per 500 windows. The composite score falls slightly, from -0.463 to -0.480 , because the fixed playbook filters less aggressively and leakage rises. The same operator also helps Sentinel (from 6.9 to 0.7), so the playbook narrows the difference between the two defenders rather than reversing it.

S11. CODE, DATA AND REPRODUCTION

The archive `IMBANG_supplementary.zip` contains:

- the simulator port, the shield, the learners and the evaluation code;
- the equivalence tests against Sentinel’s reference implementation;
- the per-seed result files behind every table and figure;
- the trained checkpoints, ten seeds for each method in Table S1 (champion and final networks, training logs, distilled rule trees and certified shield constants);
- this document.

Sentinel’s checkpoints are not redistributed; they are available from its public repository [2]. The archive’s `README.md` maps every table and figure to the script and data file that produce it. In brief:

TABLE S17
SIMULATED OPERATOR APPLYING SENTINEL’S ICMP CONSTANTS (TEN SEEDS, 20 EPISODES)

Defender	Operator (accuracy)	ICMP		ICMP-chaos		Strong		Switches
		Severe	Composite	Severe	Composite	Severe	Composite	
Sentinel	none	6.9	-0.480	28.2	-0.332	159.6	-0.753	0.0
Sentinel	ICMP playbook, 0.7	1.7	-0.481	28.2	-0.334	161.8	-0.756	4.2
Sentinel	ICMP playbook, 0.9	0.7	-0.481	28.2	-0.334	160.6	-0.754	2.2
Sentinel	ICMP playbook, 1.0	0.3	-0.482	28.1	-0.334	159.6	-0.753	1.0
IMBANG	none	17.9	-0.463	21.1	-0.286	37.9	-0.586	0.0
IMBANG	ICMP playbook, 0.7	4.2	-0.476	22.4	-0.298	47.0	-0.599	4.2
IMBANG	ICMP playbook, 0.9	2.1	-0.480	22.0	-0.298	41.2	-0.591	2.2
IMBANG	ICMP playbook, 1.0	0.5	-0.481	21.9	-0.298	37.9	-0.586	1.0

Every ten windows a simulated operator recognises the active family with the stated accuracy and, after three windows, applies protocol filtering with drop 0.5 and threshold 0.5 during ICMP floods. This study draws its own traffic, so its no-operator rows differ slightly from Table S8. Severe per 500 windows; switches: playbook changes per ICMP episode.

- `pytest` tests checks the port;
- `scripts/nl_assets.py` and `scripts/supp_assets.py` regenerate the tables and figures from the included results in under a minute;
- `scripts/evaluate.py` re-evaluates the included checkpoints under Protocols A–C;
- `scripts/train.py` retrains a method.

One training run takes 3–5 minutes on a single CPU core. The full grid of ten methods and ten seeds takes about 6–8 CPU-hours. Evaluation is deterministic: re-running Protocol B on the included checkpoints reproduces Tables S8–S9 exactly.

Scope. All results come from Sentinel’s simulator, not from production traffic. The simulator and shield port are derived from Sentinel’s MIT-licensed code [2].

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